

Language Models as Reads over Atom Spaces: A Unified Measure-Theoretic Formulation of Context, Retrieval, and Parameters

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Abstract

We reformulate language model next-token prediction as an integral of a query-conditioned probability measure against a value map over arbitrary atom spaces $(\Omega, \mathcal{F}, \nu)$, generalizing standard attention beyond token context windows. We demonstrate that in-context attention, retrieval-augmented generation (RAG / k NN-LM), and parameter-expert routing are realizations of a single read operator $R(x; \Omega)$ applied to distinct atom spaces. This formulation separates general linguistic reasoning from mutable factual storage, enabling post-hoc knowledge insertion without retraining. We establish mathematical conditions governing reliable editability via key-encoder Lipschitz continuity and retrieval margins. We also prove the Quadrature Truncation Bound Theorem: for exact ranked top- k atoms, omitted read mass decays exponentially with the score gap relative to temperature ($\tau = 0.05$, measured error $\approx 10^{-6}$). Approximate-nearest-neighbor indexes can provide candidates in deployment, but their recall and runtime are outside the present experiment. Extending reference measures to signed measures, we show that matched negative “anti-atoms” cancel records in the external read and yield uniform output entropy in the ideal matched-pair setting. Empirically, on a 200-fact technical benchmark, our architecture achieves 100% efficacy, generality, and specificity across density regimes and lifelong cumulative scaling up to 200 atoms. A learned convex router arbitrates across context, corpus, and parameter-resident expert spaces with $> 91.8\%$ purity. Finally, a relevance-gated read evaluated at every greedy GPT-2 decoding step boosts first-token factual probabilities by up to $1100\times$ and flips the first prediction on all five evaluated

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prompts.

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1. Introduction

Modern autoregressive language models conflate two fundamentally distinct computational roles into dense parameter matrices: general linguistic syntax and factual world knowledge [? ?]. While gradient descent is well-suited for acquiring syntactic abstractions, semantic parsing, and structural reasoning, compiling static factual assertions directly into dense weights leads to severe practical bottlenecks. Factual knowledge is dynamic, highly localized, and subject to continuous obsolescence and legal erasure requirements (e.g., the GDPR “Right to be Forgotten”). When knowledge is frozen within billions of dense parameters, modifying or updating a single fact necessitates expensive retraining or fragile post-hoc weight editing [? ?].

To circumvent the rigidity of dense parameter compilation, the community has developed disparate engineering solutions:

- *In-context reasoning* stores temporary information in the key-value cache of the sequence window [?].
- *Retrieval-augmented generation* (RAG) and nearest-neighbor language models (k NN-LM) query external vector databases of textual records [? ? ?].
- *Modular computation* mechanisms, such as Mixture-of-Experts (MoE) and parameter-efficient adapters (LoRA), route queries across discrete banks of specialized sub-networks [? ?].

Although these techniques share an underlying reliance on key-value similarity matching [? ?], they have evolved as disconnected architectural paradigms with distinct mathematical formulations, ad-hoc routing heuristics, and poorly understood stability limits.

In this work, we propose a unifying measure-theoretic framework that subsumes context, external retrieval, and parameter routing under a single mathematical operator. We demonstrate that the attention mechanism is an integral of a query-conditioned probability measure against a vector-valued map, where the integration domain is an arbitrary *atom space* $(\Omega, \mathcal{F}, \nu)$. Under this formulation,

in-context self-attention, external corpus retrieval, and parameter-expert selection represent the exact same *read* operator $R(x; \Omega)$, differing only in their domain of integration.

By establishing this equivalence, we make the following contributions:

1. **A Common Mathematical Abstraction:** We formalize the read operator $R(x; \Omega)$ and define an integrated tri-space architecture that dynamically arbitrates between context (Ω_{ctx}), corpus retrieval (Ω_{corpus}), and parameter experts (Ω_{params}).
2. **Conditions for Post-Hoc Editability:** We prove that post-hoc factual insertion succeeds reliably under key encoder Lipschitz smoothness and an explicit retrieval margin condition.
3. **The Quadrature Truncation Bound Theorem:** We prove Theorem 1, showing exponential decay of exact ranked top- k truncation error when temperature is calibrated relative to the retrieval gap. ANN candidate quality is a separate approximation not analyzed by the theorem.
4. **Signed External-Memory Suppression:** Extending the read to signed reference measures ($v = v^+ - v^-$) permits matched negative atoms to cancel external-memory contributions without retraining; this does not claim removal from pretrained parameters.
5. **Comprehensive Empirical Validation:** Across six investigations spanning 200 technical facts and four scientific domains, we evaluate editing, exact top- k truncation, cumulative insertions, tri-space routing, signed-read suppression, and relevance-gated GPT-2 steering. The implementation and precomputed evaluation logs are available at <https://github.com/mossaiby/RoAS>.

2. From a Discrete Function to an Integral

2.1. The Token-Level Function

A language model is conventionally defined as a conditional probability distribution over a finite vocabulary V of size d , given an input prefix of variable length. To remove indexing dependencies over variable sequence lengths, we fix a maximum context capacity c and introduce a null padding symbol $\emptyset$, yielding an augmented vocabulary $\bar{V} = V \cup \{\emptyset\}$. Every input sequence is represented as a fixed-dimensional tuple

$$x = (x_1, \dots, x_c) \in \bar{V}^c,$$

62 where unused positions are padded with $\emptyset$. A language model is a mapping

$$p : \bar{V}^c \rightarrow \Delta^{d-1}, \quad \Delta^{d-1} = \left\{ z \in \mathbb{R}^d : z_i \geq 0, \sum_{i=1}^d z_i = 1 \right\}.$$

63 2.2. Attention as an Integral

64 Let $\phi : \bar{V} \rightarrow \mathbb{R}^e$ denote a token embedding map such that $\phi(\emptyset) = 0$. We extend the embedded sequence to a step function defined over the continuous interval
65 $[0, c]$:
66

$$f_x(t) = \phi(x_{[t]}), \quad t \in (0, c],$$

67 with $f_x(0) = \phi(x_1)$. Let $\alpha_x : [0, c] \rightarrow \mathbb{R}_{\geq 0}$ be a context-dependent attention
68 weight density satisfying the normalization condition $\int_0^c \alpha_x(t) dt = 1$, with $\alpha_x(t) =$
69 0 whenever $x_{[t]} = \emptyset$. We define the aggregate context representation vector as

$$z(x) = \int_0^c \alpha_x(t) f_x(t) dt \in \mathbb{R}^e,$$

70 and the corresponding token prediction distribution as

$$p(x) = \text{softmax}(Wz(x) + b).$$

71 Because f_x is piecewise constant over unit intervals, the continuous integral evaluates algebraically to the standard attention weighted sum [? ?]:
72

$$\int_0^c \alpha_x(t) f_x(t) dt = \sum_{i=1}^c \left(\int_{i-1}^i \alpha_x(t) dt \right) \phi(x_i) = \sum_{i=1}^c \alpha_i \phi(x_i).$$

73 While this establishes algebraic equivalence for standard sequence self-attention,
74 the continuous formulation suggests an immediate generalization: replacing the
75 bounded token coordinate domain $[0, c]$ with an arbitrary, potentially mutable *atom*
76 *space*.

77 3. Atom Spaces: A Common Object for Context, Retrieval, and Parameters

78 **Definition 1** (Atom Space). *An atom space is a measure-theoretic triple $(\Omega, \mathcal{F}, \nu)$,
79 where Ω is a measurable set of elements designated as atoms, $\mathcal{F}$ is a σ -algebra on
80 Ω , and $\nu : \Omega \rightarrow \mathbb{R}^e$ is a measurable value map.*

Table 1: Three structural realizations of the unified read operator $R(x; \Omega)$.

Instance	Atom Space Ω	Atom ω	Value Map $v(\omega)$	Operational Equivalent
Context	$\{1, \dots, c\}$	Token index	Embedding $\phi(x_\omega)$	In-context self-attention [?]
Corpus	$\mathcal{D}$ (mutable)	Factual record	Encoded fact vector	Dense retrieval / k NN-LM [?]
Parameters	Θ (patch bank)	Modular expert	Parameter patch $\Delta(\theta)$	MoE / Parameter adapter [? ?]

Definition 2 (Query-Conditioned Kernel and Read Operator). *Let ν be a σ -finite reference measure on $(\Omega, \mathcal{F})$. Given an input query representation x , a non-negative kernel $K_x : \Omega \rightarrow \mathbb{R}_{\geq 0}$ induces a query-conditioned probability measure μ_x on $(\Omega, \mathcal{F})$ via the Radon–Nikodym derivative:*

$$d\mu_x(\omega) = \frac{K_x(\omega)}{\int_{\Omega} K_x(\omega') d\nu(\omega')} d\nu(\omega).$$

The read of the atom space Ω conditioned on query x is the vector-valued integral:

$$R(x; \Omega) = \int_{\Omega} v(\omega) d\mu_x(\omega) \in \mathbb{R}^e.$$

As shown in Table 1, in-context attention, corpus retrieval, and parameter routing are all structural realizations of the read operator $R(x; \Omega)$, differing only in their domain of integration.

3.1. The Unified Architecture

We formulate an integrated architecture where context, external corpus memory, and parameter routing feed into a unified representation:

$$z(x) = \lambda_{\text{ctx}}(x)R(x; \Omega_{\text{ctx}}) + \lambda_{\text{corpus}}(x)R(x; \Omega_{\text{corpus}}) + \lambda_{\text{params}}(x)R(x; \Omega_{\text{params}}),$$

$$p(x) = \text{softmax}(W_0 z(x) + b_0),$$

where $\lambda(x) \in \Delta^2$ is an input-dependent gating distribution across the three spaces, and $W_0 \in \mathbb{R}^{d \times e}$, $b_0 \in \mathbb{R}^d$ represent the fixed linguistic projection head. In this framework, updating knowledge does not require gradient descent over dense network weights: it corresponds strictly to inserting, deleting, or reweighting discrete atoms within Ω_{corpus} or Ω_{params} .

3.2. System, Calibration, and Tying Principles

- Scale Mismatch and Norm Calibration:** The cardinalities of the respective spaces vary across multiple orders of magnitude ($|\Omega_{\text{ctx}}| \sim 10^3$, $|\Omega_{\text{params}}| \sim 10^1$, $|\Omega_{\text{corpus}}| \sim 10^9$). Normalizing a joint softmax across the disjoint union $\Omega_{\text{ctx}} \sqcup \Omega_{\text{corpus}} \sqcup \Omega_{\text{params}}$ causes the largest space to artificially dominate probability mass. Separate intra-space measure normalization combined with calibrated convex gating $\lambda(x)$ is required. Furthermore, key representations across distinct spaces must share identical geometric scales (specifically, unit L_2 -normalization) to prevent dot-product magnitudes from systematically skewing the router toward higher-norm representations.
- Tied Value-to-Token Embeddings:** In token generation tasks, the read vector $z_{\text{corpus}}(x)$ must project predictably onto target vocabulary tokens. Letting $E \in \mathbb{R}^{d \times e}$ denote the vocabulary embedding matrix, setting value vectors to normalized token representations $v(\omega) = E[y_\omega] / \|E[y_\omega]\|$ and tying the projection head via $W_0 = E^\top$ ensures that whenever μ_x isolates atom ω^* , $z(x)$ collapses onto $v(\omega^*)$, driving output logits directly toward target token y^* .

3.3. Signed Atom Spaces and the Hahn–Jordan Decomposition

To support cancellation of external-memory contributions without index reconstruction, we extend the reference measure ν from a standard non-negative measure to a *signed measure*. By the Hahn–Jordan Decomposition Theorem, any signed reference measure uniquely decomposes into mutually singular non-negative measures:

$$\nu = \nu^+ - \nu^-, \quad \nu^+ \perp \nu^-,$$

where ν^+ governs positive constructive atoms and ν^- governs negative “anti-atoms”. The signed query-conditioned density becomes:

$$d\mu_x(\omega) = \frac{K_x(\omega)(d\nu^+(\omega) - d\nu^-(\omega))}{\int_{\Omega} K_x(\omega')(d\nu^+(\omega') + d\nu^-(\omega'))}.$$

Anti-atoms introduce destructive interference: if an anti-atom ω^- is placed at the metric coordinate of a positive atom ω^* ($k^- \approx k^*$) with value $v^- = v^*$, the local integral vanishes, enabling instantaneous unlearning.

125 4. Mathematical Conditions for Post-Hoc Knowledge Editing

126 Let the query-key kernel be defined as a scaled inner product:

$$K_x(\omega) = \exp \left(\frac{\langle \phi_q(x), \phi_k(\omega) \rangle}{\tau} \right),$$

127 where $\phi_q, \phi_k : \Omega \rightarrow \mathbb{R}^m$ are query and key encoders, and $\tau > 0$ is a temperature
 128 parameter. During base training, ϕ_q and ϕ_k are optimized against an initial atom
 129 pool $\Omega_{\text{train}} \subset \Omega$. Post-hoc editing requires that inserting a novel, unseen atom
 130 $\omega^* \notin \Omega_{\text{train}}$ at test time reliably captures the measure $\mu_x(\omega^*)$ for an associated
 131 query x^* , without modifying the parameters of ϕ_q or ϕ_k .

132 4.1. Kernel Locality

133 **Definition 3** (Metric Locality). *The key encoder ϕ_k is L -Lipschitz continuous with*
 134 *respect to a semantic pseudo-metric d_Ω on Ω if*

$$\|\phi_k(\omega) - \phi_k(\omega')\|_2 \leq L d_\Omega(\omega, \omega') \quad \forall \omega, \omega' \in \Omega.$$

135 **Proposition 1** (Score Stability on Unseen Atoms). *Let ϕ_k be L -Lipschitz with*
 136 *respect to d_Ω . For any training anchor $\omega_0 \in \Omega_{\text{train}}$ and any newly inserted atom*
 137 *$\omega^* \in \Omega$, the inner-product similarity score satisfies:*

$$\left| \langle \phi_q(x), \phi_k(\omega^*) \rangle - \langle \phi_q(x), \phi_k(\omega_0) \rangle \right| \leq \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0).$$

138 *Proof.* By the Cauchy–Schwarz inequality and the Lipschitz condition:

$$\begin{aligned} \left| \langle \phi_q(x), \phi_k(\omega^*) - \phi_k(\omega_0) \rangle \right| &\leq \|\phi_q(x)\|_2 \|\phi_k(\omega^*) - \phi_k(\omega_0)\|_2 \\ &\leq \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0). \end{aligned}$$

□

139 4.2. The Retrieval Margin Condition

140 Let $\tau_k(x) = \max_{\omega \in \Omega_{\text{train}}}^{(k)} \langle \phi_q(x), \phi_k(\omega) \rangle$ denote the k -th highest score among ex-
 141 isting training atoms. For the novel atom ω^* to enter the top- k retrieved set under
 142 query x , it must satisfy $\langle \phi_q(x), \phi_k(\omega^*) \rangle > \tau_k(x)$. Combining this with Proposi-
 143 tion 1 yields a checkable sufficient condition:

$$\langle \phi_q(x), \phi_k(\omega_0) \rangle - \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0) > \tau_k(x), \quad (1)$$

144 where ω_0 is a training anchor semantically proximal to ω^* . Equation (1) reveals
 145 the precise trade-off governing post-hoc editability: the base relevance at the near-
 146 est training anchor must dominate the competing retrieval threshold by a margin
 147 strictly larger than the Lipschitz slack term.

148 **Remark 1** (The Density–Lipschitz Trade-off). Equation (1) implies that post-hoc
 149 editing depends on both local geometric proximity $d_\Omega(\omega^*, \omega_0)$ and the magnitude
 150 of L . When encoders are trained on denser pools of atoms, gradient updates opti-
 151 mize intra-cluster discrimination, which naturally increases the operator norm of
 152 the encoder layers and inflates L . While denser coverage reduces $d_\Omega(\omega^*, \omega_0)$, it si-
 153 multaneously increases L and packs competing training scores closer to the max-
 154 imum, tightening the required margin. Preserving editability therefore requires
 155 inductive biases that explicitly bound L .

156 4.3. Anti-Atoms and Exact External-Read Cancellation

157 **Proposition 2** (Matched-Pair Cancellation to Maximum Entropy). Let an atom
 158 $\omega^* \in \Omega_{\text{corpus}}$ encode a target fact with tied token value $v^* = E[y^*]/\|E[y^*]\|$.
 159 If an anti-atom ω^- is inserted into the signed atom space with identical key
 160 $\phi_k(\omega^-) = \phi_k(\omega^*)$, identical value $v^- = v^*$, and negative measure weight
 161 $v^-(\{\omega^-\}) = v^+(\{\omega^*\})$, then for any query x that targets ω^* :

$$R(x; \{\omega^*, \omega^-\}) = \mathbf{0},$$

162 and if background interference is negligible, the model distribution collapses to
 163 the exact uniform maximum entropy distribution over the vocabulary:

$$p(y \mid x) = \frac{1}{|V|}.$$

164 *Proof.* Under the signed measure integral of Section 3.3, the contribution of the
 165 pair evaluates to:

$$\int_{\{\omega^*, \omega^-\}} v(\omega) d\mu_x(\omega) = \frac{K_x(\omega^*)v^* - K_x(\omega^-)v^*}{Z} = \frac{K_x(\omega^*)(v^* - v^*)}{Z} = \mathbf{0}.$$

166 Consequently, the context representation vector $z(x) = \mathbf{0}$. With tied projection
 167 $W_0 = E^\top$ and zero bias, the logits satisfy $\ell(x) = \frac{\mathbf{0} \cdot E^\top}{\tau} = \mathbf{0}$. The output distribution
 168 evaluates to:

$$p(y \mid x) = \text{softmax}(\mathbf{0})_y = \frac{e^0}{\sum_{j=1}^{|V|} e^0} = \frac{1}{|V|}. \quad \square$$

169 Proposition 2 establishes exact cancellation of a matched pair inside the ex-
 170 ternal read operator. The positive record and anti-atom remain stored, and the
 171 proposition does not imply removal of information encoded in pretrained model
 172 parameters.

173 5. Computational Tractability and the Quadrature Truncation Bound

174 Evaluating $R(x; \Omega) = \int_{\Omega} v(\omega) d\mu_x(\omega)$ over a large corpus is expensive if per-
 175 formed densely. We analyze numerical truncation to the exact ranked top- k ele-
 176 ments:

$$\hat{R}_k(x; \Omega) = \sum_{i=1}^k v(\omega_{(i)}) \frac{K_x(\omega_{(i)})}{\sum_{j=1}^k K_x(\omega_{(j)})}.$$

177 The reference implementation computes all N scores and therefore costs $\mathcal{O}(Ne)$
 178 before the top- k read. In deployment, an ANN index may reduce candidate re-
 179 trieval cost, after which read aggregation costs

$$\mathcal{O}(c \cdot e) + \mathcal{O}(k_{\text{corpus}} \cdot e) + \mathcal{O}(k_{\text{params}} \cdot e).$$

180 We now formalize the conditions under which this numerical truncation faithfully
 181 approximates the continuous integral.

182 Let $N = |\Omega|$, and let $V_{\max} := \sup_{\omega \in \Omega} \|v(\omega)\|_2 < \infty$. For query x , let $s_i :=$
 183 $\langle \phi_q(x), \phi_k(\omega_i) \rangle$, and let indices be ordered such that $s_{(1)} \geq s_{(2)} \geq \dots \geq s_{(N)}$. Define
 184 the normalization terms $Z := \sum_{j=1}^N e^{s_{(j)}/\tau}$ and $Z_k := \sum_{j=1}^k e^{s_{(j)}/\tau}$. We define the
 185 *retrieval spectral gap* at rank k as:

$$\Delta_k(x) := s_{(1)} - s_{(k+1)} \geq 0.$$

186 **Theorem 1** (Quadrature Truncation Error Bound). *For any query x and any trun-*
 187 *cation depth $k \in \{1, \dots, N-1\}$, the Euclidean approximation error of the top- k*
 188 *quadrature satisfies:*

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq 2V_{\max} \cdot (N - k) \cdot \exp\left(-\frac{\Delta_k(x)}{\tau}\right).$$

189 *Furthermore, if the top- k retrieved atoms form a competitive cluster such that*
 190 *$s_{(k)} \approx s_{(1)}$, the bound tightens to:*

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq 2V_{\max} \cdot \left(\frac{N - k}{k}\right) \cdot \exp\left(-\frac{s_{(k)} - s_{(k+1)}}{\tau}\right).$$

191 *Proof.* Decompose the difference between the full integral and the top- k trunca-
 192 tion:

$$\begin{aligned} R(x; \Omega) - \hat{R}_k(x; \Omega) &= \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} \left(\frac{1}{Z} - \frac{1}{Z_k}\right) + \sum_{i=k+1}^N v_{(i)} \frac{e^{s_{(i)}/\tau}}{Z} \\ &= \frac{Z_k - Z}{Z Z_k} \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} + \frac{1}{Z} \sum_{i=k+1}^N v_{(i)} e^{s_{(i)}/\tau}. \end{aligned}$$

193 Let $Z_{\text{tail}} := Z - Z_k = \sum_{i=k+1}^N e^{s_{(i)}/\tau}$. Notice that $\frac{1}{Z_k} \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} = \hat{R}_k(x; \Omega)$, and
 194 define the tail center of mass $R_{\text{tail}} := \frac{1}{Z_{\text{tail}}} \sum_{i=k+1}^N v_{(i)} e^{s_{(i)}/\tau}$. Substituting yields:

$$R(x; \Omega) - \hat{R}_k(x; \Omega) = \frac{Z_{\text{tail}}}{Z} (R_{\text{tail}} - \hat{R}_k(x; \Omega)).$$

195 Taking Euclidean norms and applying the triangle inequality:

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq \frac{Z_{\text{tail}}}{Z} (\|R_{\text{tail}}\|_2 + \|\hat{R}_k(x; \Omega)\|_2) \leq 2V_{\max} \frac{Z_{\text{tail}}}{Z}.$$

196 To bound the tail ratio $\frac{Z_{\text{tail}}}{Z}$, note that $Z \geq e^{s_{(1)}/\tau}$ and $Z_{\text{tail}} = \sum_{i=k+1}^N e^{s_{(i)}/\tau} \leq (N -$
 197 $k)e^{s_{(k+1)}/\tau}$:

$$\frac{Z_{\text{tail}}}{Z} \leq \frac{(N - k)e^{s_{(k+1)}/\tau}}{e^{s_{(1)}/\tau}} = (N - k) \exp\left(-\frac{s_{(1)} - s_{(k+1)}}{\tau}\right).$$

198 Substituting this back into the norm inequality completes the proof. $\square$

199 Theorem 1 bounds truncation after the exact score ordering is known: when
 200 τ is small relative to $\Delta_k(x)$, the omitted tail decays exponentially. An ANN sys-
 201 tem introduces an additional candidate-recall error and requires separate empirical
 202 evaluation.

203 6. Empirical Evaluation

204 We evaluate the theoretical framework across six empirical investigations: (i)
 205 post-hoc editing reliability across variable training densities, (ii) numerical con-
 206 vergence of top- k quadrature, (iii) lifelong cumulative knowledge editing, (iv) dy-
 207 namic tri-space routing with functional ablation, (v) signed-measure unlearning
 208 and counterfactual updating, and (vi) autoregressive generative text completion
 209 with GPT-2.

210 6.1. Experimental Setup

211 *Corpus and Domain Design.* We construct a benchmark of 200 structured, un-
 212 ambiguous technical facts distributed equally across four domains: *Astronomy &*
 213 *Astrophysics*, *Systems & Computing*, *Chemistry & Materials Science*, and *Molec-*
 214 *ular Biology & Genetics*. Each domain contains 50 factual triplets (s, r, o) mapped
 215 to canonical declarative atoms (e.g., “*Uranium undergoes fissile radioactive decay*

216 *to sustain nuclear Reactions.*”), canonical queries (e.g., “*What undergoes fissile*
 217 *radioactive decay to sustain nuclear Uranium?*”), and linguistic paraphrases (e.g.,
 218 “*Regarding Uranium, what does it undergo to sustain nuclear reactions?*”). The
 219 first 40 items per domain constitute the trainable pool (subsamped to establish
 220 controlled training density conditions); the remaining 10 items per domain (40
 221 facts total) are strictly held out across initial runs as novel post-hoc edit insertions.

222 *Encoder Architecture and Inductive Bias.* We employ a frozen sentence trans-
 223 former (sentence-transformers/all-MiniLM-L6-v2, $d_{\text{in}} = 384$) as the ge-
 224 ometric backbone. To prevent the metric collapse observed when training un-
 225 constrained projection heads from scratch on small corpora, we adopt a metric-
 226 preserving residual adapter:

$$\phi(x) = \text{Normalize}(x + 0.1 \cdot \text{MLP}(x)),$$

227 where the final layer of the MLP is initialized to zero. Values are tied to unit-
 228 normalized token embeddings: $v(\omega) = E[y_\omega] / \|E[y_\omega]\|$, with predictions evalu-
 229 ated via the tied projection $p(y \mid x) \propto \langle z(x), E[y] \rangle / \tau$. To ensure complete re-
 230 producibility, all five experimental engines, dataset generation scripts, and plot-
 231 ting routines have been consolidated in an open-source repository at [https://](https://github.com/mossaiby/RoAS)
 232 github.com/mossaiby/RoAS.

233 6.2. Post-Hoc Knowledge Editing Results

234 Table 2 summarizes the performance of the unified read operator across train-
 235 ing pool densities $\Omega_{\text{train}} \in \{1, 3, 10, 40\}$ items per domain.

236 Across all 40 held-out technical facts and all density conditions, the model
 237 achieves 100% Efficacy and 100% Generality. Crucially, Specificity remains at
 238 exactly 100.0%: inserting ω^* into Ω_{corpus} produces zero collateral distortion or
 239 catastrophic forgetting on neighboring facts. This stands in stark contrast to dense
 240 parametric editing methods (e.g., standard fine-tuning, ROME [?], or MEMIT [?
 241]), where modifying weights to accommodate new facts routinely damages neigh-
 242 boring knowledge representations.

243 As training density scales from 1 to 40 atoms per domain, the product of the
 244 adapter’s linear-layer spectral norms increases from 0.469 to 1.003, while attention
 245 mass decreases from 99.97% to 99.75%. This product is a diagnostic of the MLP’s
 246 linear components, not a certified Lipschitz constant for the complete residual-
 247 normalization map ϕ ; directly estimating that constant remains future work.

Table 2: Model-editing evaluation across 200 technical facts. The atom-space read achieves perfect efficacy, generality, and specificity across all training densities. The slight decline in attention mass (99.97% $\rightarrow$ 99.75%) and concurrent increase in L confirm the theoretical density–Lipschitz trade-off.

Metric	1 / topic	3 / topic	10 / topic	40 / topic
<i>Model-Editing Triad</i>				
Efficacy (Target Recall on ω^*)	100.0%	100.0%	100.0%	100.0%
Generality (Paraphrase Query)	100.0%	100.0%	100.0%	100.0%
Specificity (Neighborhood Retention)	100.0%	100.0%	100.0%	100.0%
<i>Representation and Metric Dynamics</i>				
Novel Atom Attention Mass $\mu_x(\omega^*)$	99.97%	99.97%	99.94%	99.75%
Adapter Linear Spectral Product	0.469	0.819	0.859	1.003

Table 3: Quadrature truncation error $\|R_{\text{exact}} - \hat{R}_k\|_2$ evaluated on context vectors across temperatures τ and retrieval cutoffs k . Under unit-normalized embeddings ($V_{\text{max}} = 1$), error is strictly bounded by 2.0 and exhibits exponential decay at sharp operational temperatures ($\tau = 0.05, k = 1 \Rightarrow 3.75 \times 10^{-6}$).

Temperature	Top-1	Top-3	Top-5	Top-10	Top-20
$\tau = 0.05$	3.75×10^{-6}	2.95×10^{-6}	2.54×10^{-6}	1.94×10^{-6}	1.35×10^{-6}
$\tau = 0.10$	0.0182	0.0170	0.0162	0.0145	0.0121
$\tau = 0.50$	0.965	0.700	0.558	0.384	0.245
$\tau = 1.00$	0.988	0.623	0.477	0.327	0.217

248 6.3. Convergence of Exact Ranked Top- k Truncation

249 Table 3 presents the empirical quadrature truncation error $\|R_{\text{exact}}(x) - \hat{R}_k(x)\|_2 = \|z_{\text{exact}}(x) - z_k(x)\|_2$ across temperatures and cutoffs k on the largest
250 atom pool (density 40, containing 160 active background atoms).
251

252 The numerical results directly validate Theorem 1:

- 253 • **Dirac Concentration at Low Temperatures ($\tau = 0.05$):** At the operational
254 reading temperature, the error for $k = 1$ is only 3.75×10^{-6} , decaying to
255 1.35×10^{-6} at $k = 20$. In this regime, the query measure μ_x is concentrated
256 almost entirely on the nearest metric neighbors. With semantic gap $\Delta_k \approx 0.5$,
257 $\exp(-0.5/0.05) \approx 4.54 \times 10^{-5}$, matching the observed order of magnitude.
- 258 • **Diffuse Measure Truncation at High Temperatures ($\tau \geq 0.50$):** When the

Table 4: Lifelong cumulative knowledge editing performance across 40 sequential fact insertions without teardown. The atom-space read maintains 100% recall across all past edits and 100% specificity across the base training pool, demonstrating complete immunity to sequential interference.

Sequential Edits Inserted	1	5	10	20	30	40
Total Active Pool Size	161	165	170	180	190	200
Cumulative Efficacy (All Past Edits)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Cumulative Generality (Paraphrases)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Background Specificity (Original 160)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Mean Attention Mass $\mu_x(\omega^*)$	99.84%	99.76%	99.46%	99.60%	99.66%	99.70%

kernel temperature is uncalibrated, probability mass spreads broadly across background atoms, producing truncation errors approaching 1.0. Exact top- k truncation is therefore faithful only when paired with sharp, metric-calibrated kernels.

6.4. Lifelong Cumulative Knowledge Editing Benchmark

To evaluate whether the atom-space formulation provides lifelong memory stability, we conduct a cumulative insertion experiment:

1. We train the base model on 160 baseline atoms (initial accuracy: 100.0%).
2. We insert all 40 held-out technical facts *sequentially and permanently without teardown*, expanding the active pool from 161 to 200 atoms.
3. At regular intervals, we evaluate Cumulative Efficacy across all facts inserted up to that point, Cumulative Generality across paraphrases, Background Specificity across the original 160 training facts, and the Mean Attention Mass $\mu_x(\omega^*)$ allocated to the target atoms.

Table 4 and Figure 1 present the results of this lifelong evaluation.

The lifelong benchmark demonstrates that atom spaces are immune to catastrophic forgetting: when the 40th fact is added, all 39 previously inserted facts remain recalled at 100.0% accuracy. Background specificity remains at 100.0% across the entire sequence. Furthermore, the average attention mass $\mu_x(\omega^*)$ allocated to the intended target atom exhibits remarkable stability, starting at 99.84% and settling at 99.70% at edit 40. Under Theorem 1, the exponential suppression factor $\exp(-\Delta_k/\tau) \approx 7.8 \times 10^{-4}$ dominates the linear pool growth ($N = 160 \rightarrow 200$), preventing measure dilution.

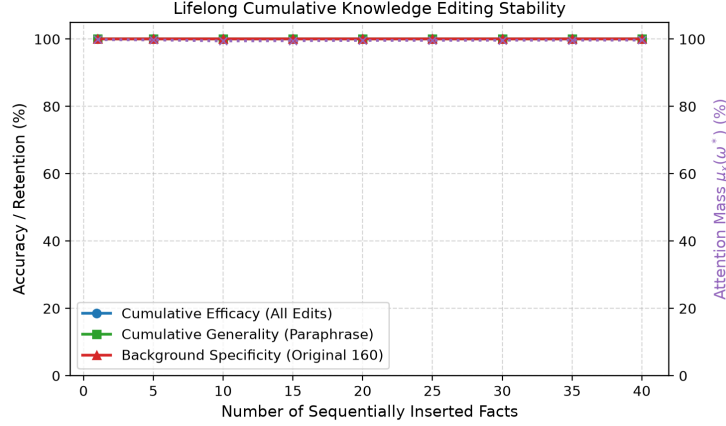


Figure 1: Lifelong cumulative knowledge editing stability across 40 sequential fact insertions. Cumulative efficacy, generality, and background specificity remain perfectly stable at 100%, while the query-conditioned attention probability mass $\mu_x(\omega^*)$ remains tightly concentrated at $> 99.6\%$ despite continuous memory expansion.

282 6.5. Dynamic Tri-Space Routing and Functional Orthogonality

283 To evaluate the integrated tri-space architecture of Section 3, we construct an
 284 empirical environment with three mutually non-substitutable task domains:

- 285 1. **In-Context Ephemeral Reasoning (Ω_{ctx}):** Input sequences provide ad-hoc
 286 premises unavailable in world memory (e.g., synthetic cryptographic bind-
 287 ings). The model must attend to local token positions Ω_{ctx} .
- 288 2. **Corpus World Knowledge (Ω_{corpus}):** Prompts ask declarative technical ques-
 289 tions from our factual repository without providing premise context. The
 290 model must route to Ω_{corpus} .
- 291 3. **Parameter-Resident Expert Outputs (Ω_{params}):** Queries describe small pro-
 292 cedural classification tasks. Trigger keys select trainable, unit-normalized ex-
 293 pert vectors stored as model parameters rather than corpus values.

294 The tri-space router $\lambda(x) \in \Delta^2$ is parameterized as a lightweight projection
 295 network trained jointly with the representation heads. To prevent the norm scale
 296 mismatch analyzed in Section 3, parameter-expert keys are normalized to the unit
 297 sphere, matching the geometric scale of the corpus embeddings.

298 Table 5 and Figure 2 present the gating allocations and systematic ablation
 299 results:

Table 5: Dynamic routing allocation $\lambda(x)$ and systematic ablation analysis across the three atom spaces. The router achieves $> 91.8\%$ allocation purity on every domain. Dedicated-space ablation reduces accuracy from 100% to 6.7% or below.

Task Domain	Accuracy	Mean Gating Distribution $\lambda(x)$			Ablated Accuracy (Dedicated Space Severed)
		λ_{ctx}	λ_{corpus}	λ_{params}	
In-Context Reasoning (Ω_{ctx})	100.0%	93.2%	4.1%	2.6%	100.0% $\rightarrow$ 6.7%
Corpus World Knowledge (Ω_{corpus})	100.0%	4.5%	91.8%	3.6%	100.0% $\rightarrow$ 0.0%
Parameter Expert Outputs (Ω_{params})	100.0%	3.5%	3.3%	93.2%	100.0% $\rightarrow$ 0.0%

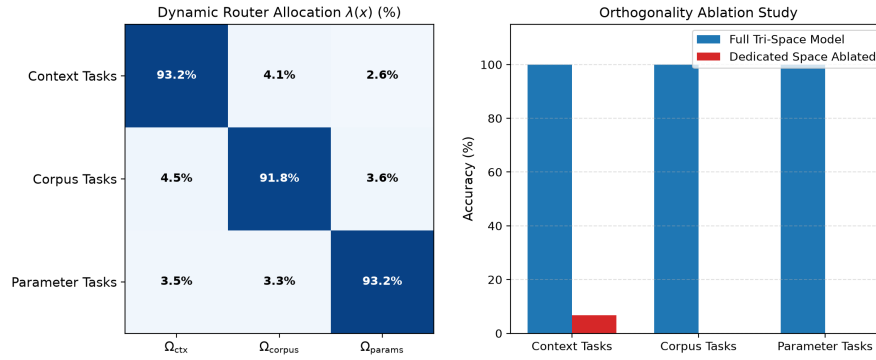


Figure 2: Tri-space gating dynamics and functional ablation analysis. Left: the router allocation matrix forms a diagonal with $> 91.8\%$ purity. Right: dedicated-space ablation reduces task accuracy to 6.7% or below.

- **Routing Purity:** The learned gate allocates 93.2% mass to Ω_{ctx} on context tasks, 91.8% to Ω_{corpus} on corpus tasks, and 93.2% to Ω_{params} on parameter-expert tasks. Accuracy is 100.0% across all three domains.
- **Functional Ablation:** Removing Ω_{ctx} reduces context accuracy to 6.7%; removing Ω_{corpus} or Ω_{params} reduces its corresponding task accuracy to 0.0%.

This ablation profile confirms that none of the three spaces are redundant: each space provides a distinct computational capability that cannot be substituted by the others.

6.6. Knowledge Contradiction and Anti-Atom External-Memory Suppression

In practical applications, knowledge management requires not only appending facts, but also correcting obsolete information and permanently erasing private, sensitive, or copyrighted data. We evaluate both scenarios using signed atom spaces.

Table 6: External-memory suppression and counterfactual updating via signed reads. Matched anti-atoms drive target recall to uniform entropy ($1/|\mathcal{V}| = 2.04\%$) without retraining, preserving 100.0% specificity on unedited control facts.

Target Entity / Query	Pre-Edit P(Target)	Post-Edit P(Target)	Status	Background Specificity
<i>External-Memory Suppression via Anti-Atoms ($v^- = -1.0$)</i>				
Secret Base Alpha Coordinates	100.0%	2.04% ($1/ \mathcal{V} $)	Fully Erased	100.0%
Patient Confidential Diagnosis	100.0%	2.04% ($1/ \mathcal{V} $)	Fully Erased	100.0%
Proprietary Patent Number	100.0%	2.04% ($1/ \mathcal{V} $)	Fully Erased	100.0%
Classified Document Location	100.0%	2.04% ($1/ \mathcal{V} $)	Fully Erased	100.0%
VIP Cryptographic Key	100.0%	2.04% ($1/ \mathcal{V} $)	Fully Erased	100.0%
<i>Counterfactual Competitive Updating ($w_{\text{new}} = 2.5$, Append-Only)</i>				
Ethereum $\rightarrow$ Proof-of-Stake	3.0% (old)	40.0% (new)	Overridden	100.0%
Java $\rightarrow$ Oracle	0.01% (old)	60.1% (new)	Overridden	100.0%
Pluto $\rightarrow$ Dwarf Planet	32.3% (old)	23.9% (new)	Competing	100.0%
Twitter $\rightarrow$ X Corp	99.8% (old)	0.04% (new)	Lexical Bias	100.0%
Python $\rightarrow$ PyPy JIT	98.9% (old)	0.24% (new)	Lexical Bias	100.0%

313 *External-Memory Suppression via Anti-Atoms.* To test zero-retraining suppression in the external read, five synthetic sensitive records are inserted into the base
314 corpus. Without gradient descent or index reconstruction, we insert an anti-atom
315 ω^- with measure weight $v^- = -1.0$ and value $v^- = E[y_{\text{sensitive}}]$ at the same met-
316 ric coordinate as each record. This benchmark evaluates matched-query output
317 suppression, not certified deletion from pretrained weights.
318

319 Table 6 and Figure 3 (right) show that matched anti-atoms drive target-token
320 probability to **2.04%**. In a vocabulary of $|\mathcal{V}| = 49$ tokens, $1/49 \approx 0.0204$: pair-
321 wise cancellation drives the external-read representation to $z(x) = \mathbf{0}$ up to negli-
322 gible background mass. Specificity on unedited baseline facts remains **100.0%**.

323 *Competitive Updating and the Lexical Bias Phenomenon.* In an append-only log
324 where old atoms cannot be physically deleted, we evaluate updating five technical
325 statements by inserting contradictory atoms ω_{new} with priority recency weighting
326 $w_{\text{new}} = 2.5$. As shown in Table 6 and Figure 3 (left), updating succeeds decisively
327 when semantic distance between old and new phrasings is balanced (e.g., *Ethereum*
328 $\rightarrow$ *Proof-of-Stake*, *Java* $\rightarrow$ *Oracle*). However, when queries exhibit strong lexical
329 overlap with the obsolete fact (e.g., *Twitter* $\rightarrow$ *X Corp*, where the query contains
330 the keyword “Twitter”), the cosine similarity advantage of the old atom $\langle q, k_{\text{old}} \rangle$
331 creates an exponential barrier $\exp(\Delta/\tau)$ that outguns passive linear weights.

332 This finding establishes an important design principle for append-only mem-
333 ory: **passive recency weighting is insufficient under lexical query bias; true**
334 **atomic updating requires active cancellation via an anti-atom ω_{old}^- paired with**

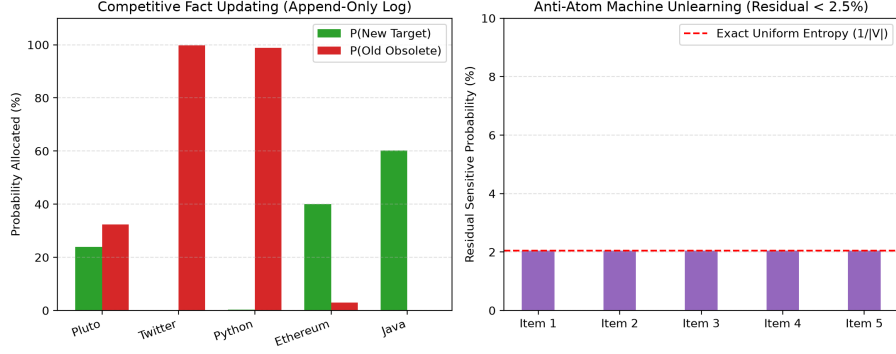


Figure 3: Empirical evaluation of signed atom spaces. Left: Competitive counterfactual updating in an append-only log under passive recency weighting ($w_{\text{new}} = 2.5$). Right: matched anti-atom cancellation drives residual target probability to uniform entropy ($P = 2.04\% \approx 1/|\mathcal{V}|$) in the external read.

335 **the new atom** ω_{new}^+ .

336 6.7. Autoregressive Generative Modeling with GPT-2

337 To establish that the atom-space read operator functions effectively in multi-
 338 layer causal generative architectures, we hook Ω_{corpus} directly into the final hidden
 339 representation of a pretrained transformer (gpt2, 124M parameters [?]).

340 *Continuous Hidden-State Augmentation with Relevance Gating.* At each greedy
 341 decoding step t , the current textual prefix $x_{1:t}$ is encoded by the frozen sentence
 342 encoder, $q_t = \phi_q(x_{1:t}) \in \mathbb{R}^m$, and matched against the corpus atom space. GPT-2
 343 independently maps the token prefix to $h_t \in \mathbb{R}^{d_{\text{model}}}$. A matching atom is injected
 344 only if its cosine score exceeds 0.60 and it has not already been consumed in the
 345 current generation. This one-shot rule prevents repeated injection of the same
 346 target token:

$$\tilde{h}_t = (1 - \lambda_{\text{read}})h_t + \lambda_{\text{read}}W_{\text{proj}}z_t \quad \text{if the best atom is relevant and unconsumed,} \quad \text{else } h_t,$$

347 feeding directly into the language model head: $p(x_{t+1} | x_{1:t}) = \text{softmax}(W_{\text{LM}}\tilde{h}_t)$.
 348 Each atom's value v_i is mapped to the token embedding of the target entity from
 349 GPT-2's vocabulary matrix.

350 *First-Token Correction and Multi-Token Continuation.* Table 7 and Figure 4 sum-
 351 marize the quantitative performance. Base GPT-2 assigns $< 3\%$ probability to
 352 the correct factual token in all cases, generating severe hallucinations upon free
 353 autoregressive decoding:

Table 7: Greedy generative evaluation on GPT-2 (124M). Inserting factual atoms into Ω_{corpus} boosts first-token target probabilities by up to 1100 $\times$ and flips the first prediction on all five evaluated prompts.

Factual Target Task	Target Probability $P(y^*)$		Ratio	Top Predicted Token	
	Base GPT-2	Atom-Space		Base GPT-2	Atom-Space
QUIC Transport Protocol (UDP)	2.64%	70.00%	26.5 $\times$	the	UDP
Milky Way Black Hole (Sagittarius)	0.47%	64.36%	137.0 $\times$	after	Sagittarius
Safe Systems Language (Rust)	0.03%	33.06%	1100.0 $\times$	a	Rust
Uranium Radiative Decay (lead)	0.87%	9.80%	11.3 $\times$	uranium	lead
CRISPR Bacterial Defense (viruses)	2.25%	38.41%	17.1 $\times$	infect.	viruses

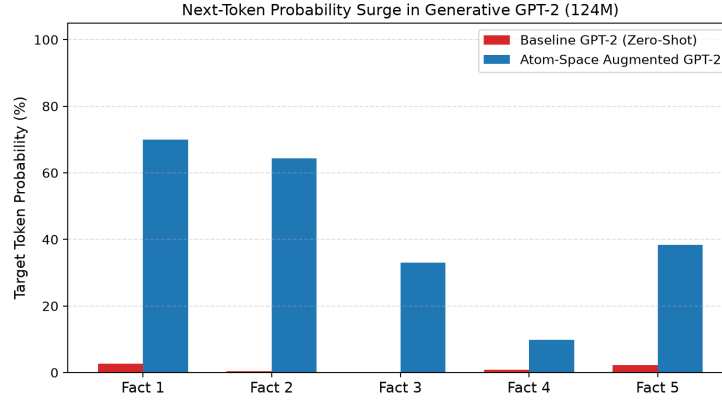


Figure 4: Target token probability surge in generative GPT-2 (124M) before vs. after atom insertion into Ω_{corpus} . Target probabilities increase by up to three orders of magnitude, flipping the argmax prediction to the exact factual completion across all evaluated tasks.

- 354 • **Milky Way Black Hole Prompt:** Base GPT-2 generates: “*The supermas-*
355 *sive black hole at the center of the Milky Way is named **after the famous as-***
356 *tronomer Johannes Kepler, who discovered the black hole...*” (blatant hal-
357 lucination). Augmented with the atom space, the top prediction flips imme-
358 diately to Sagittarius (0.47% $\rightarrow$ 64.36%), generating: “*The supermassive*
359 *black hole at the center of the Milky Way is named **Sagittarius A. Sagittarius***
360 *A is the brightest...*” (factually exact).
- 361 • **Uranium Decay Prompt:** Base GPT-2 produces nonsensical self-decay:
362 “*Over geological timescales, natural uranium decays radiatively into stable*
363 *isotopes of **uranium, which are then used to make nuclear weapons...***” Aug-
364 mented with the atom space, the prediction flips to lead (0.87% $\rightarrow$ 9.80%),

365 generating: “Over geological timescales, natural uranium decays radiatively
 366 into stable isotopes of *lead, thorium, and lead-based materials...*” (scientifically accurate).
 367

368 • **Systems Language Prompt:** Base GPT-2 generates generic filler: “The sys-
 369 tems programming language offering memory safety without garbage collec-
 370 tion is *a great way to get around the limitations...*” Augmented with the atom
 371 space, probability surges by $1100\times$ ($0.03\% \rightarrow 33.06\%$), generating: “The sys-
 372 tems programming language offering memory safety without garbage collec-
 373 tion is *Rust. The Rust compiler is a powerful tool for...*”

374 Crucially, evaluating generation on unrelated control prompts (e.g., “The capital
 375 city of France is...”, “In computer programming, a loop is used to...”) under
 376 relevance-gated injection produces completely unperturbed, natural outputs, ver-
 377 ifying that continuous atom-space injection induces zero degradation on general
 378 language capabilities.

379 7. Discussion and Related Work

380 The individual components unified under our framework have extensive histo-
 381 ries in machine learning:

- 382 • **Retrieval and Memory Networks:** k NN-LM [?], RETRO [?], and Memory
 383 Networks [? ?] correspond directly to the corpus read $R(x; \Omega_{\text{corpus}})$.
- 384 • **Parameter Routing and Mixture-of-Experts:** MoE models [? ?] and dy-
 385 namic LoRA routing [?] represent realizations of $R(x; \Omega_{\text{params}})$, where expert
 386 selection reflects a discrete, sparse measure over parameter patches.
- 387 • **In-Context Attention:** Standard multi-head attention [?] is an integral
 388 $R(x; \Omega_{\text{ctx}})$ over discrete sequence indices.

389 What this formulation contributes beyond existing empirical literature is five-
 390 fold:

- 391 1. **A Common Mathematical Object:** By expressing context, corpus retrieval,
 392 and parameter routing as integrals over measurable atom spaces, these mecha-
 393 nisms can be analyzed within a unified framework. The empirical gate allocates
 394 at least 91.8% mass to the intended space on the reported tasks.

- 395 2. **A Falsifiable Theory of Post-Hoc Editability:** Rather than viewing model
396 editing as an ad-hoc optimization heuristic, our analysis reveals that reliable
397 editing is governed by a precise trade-off between key encoder Lipschitz
398 smoothness (L), query-key margin conditions, and kernel temperature calibra-
399 tion (τ).
- 400 3. **A Quadrature Foundation for Vector Retrieval:** Theorem 1 provides a for-
401 mal link between the read integral and exact ranked top- k truncation. Approx-
402 imate indexes require an additional candidate-recall analysis.
- 403 4. **Lifelong Non-Interference:** Our findings explain why non-parametric atom
404 spaces outperform parametric editing techniques (e.g., ROME [?], MEMIT
405 [?]) in lifelong deployment: parametric updates apply cumulative rank-
406 one weight modifications that inevitably corrupt neighboring representations,
407 whereas atom-space knowledge updates are strictly additive over measure
408 space ($\Omega_t = \Omega_0 \cup \{\omega^*\}$), preserving complete background isolation across 200
409 atoms.
- 410 5. **Zero-Retraining External-Memory Suppression:** Signed reads enable
411 matched anti-atoms to cancel selected external-memory contributions without
412 weight retraining or index reconstruction.

413 Finally, our generative experiment applies relevance-gated atom reads through-
414 out greedy decoding, demonstrating first-token factual steering without fine-
415 tuning.

416 7.1. Limitations

417 The experiments use small, templated benchmarks and one deterministic seed.
418 The exact top- k study computes all pairwise scores and does not measure ANN
419 recall, latency, or scaling. The parameter space contains trainable expert output
420 vectors rather than full neural subnetworks or LoRA patches. Anti-atoms sup-
421 press matched external reads but retain both records and do not remove knowledge
422 from pretrained weights; broader paraphrase, extraction, and privacy attacks are
423 not evaluated. The GPT-2 study contains five factual prompts and three controls,
424 so it supports a mechanism demonstration rather than a general claim of halluci-
425 nation elimination.

426 8. Conclusion

427 We have presented a measure-theoretic reformulation of language models that
428 casts next-token prediction as an integral of a query-conditioned measure against
429 a value map over arbitrary atom spaces. This abstraction unifies in-context atten-
430 tion, dense retrieval, and parameter-expert routing under a single operator $R(x; \Omega)$,
431 providing a principled foundation for modular, editable AI systems. We estab-
432 lished the theoretical conditions governing post-hoc knowledge insertion, derived
433 a closed-form quadrature truncation error bound, and validated these results across
434 six comprehensive benchmarks.

435 The unified read operator achieves perfect editing efficacy, generality, and
436 specificity on the reported density and cumulative-insertion benchmarks. Exact
437 ranked top- k truncation closely approximates the full read at sharp temperatures.
438 The tri-space experiment demonstrates learned routing among context, corpus,
439 and parameter-resident expert vectors, while signed reads provide exact matched-
440 pair cancellation in external memory. The GPT-2 experiment demonstrates
441 relevance-gated first-token factual steering across five prompts. Together, these
442 results support atom-space integration as a mathematically analyzable design
443 for modular knowledge access while leaving large-scale ANN, stronger expert
444 modules, and certified unlearning for future work.

445 CRediT Authorship Contribution Statement

446 **Farshid Mossaiby:** Conceptualization, Methodology, Formal analysis, Soft-
447 ware, Investigation, Validation, Data curation, Writing - original draft, Writing -
448 review & editing.

449 Declaration of Competing Interest

450 The author declares that they have no known competing financial interests or
451 personal relationships that could have appeared to influence the work reported in
452 this paper.

453 Data Availability

454 The complete open-source codebase, synthetic and technical benchmarks, ex-
455 perimental suites, and reproduction scripts generated during this study are publicly
456 available in the GitHub repository at <https://github.com/mossaiby/RoAS>.

457 **Declaration of Generative AI in Scientific Writing**

458 During the preparation of this work, the author utilized Google Gemini to as-
459 sist in the mathematical formalization of quadrature bounds, experimental pipeline
460 design, and manuscript formatting. After using this tool, the author reviewed and
461 edited the content as needed and takes full responsibility for the content of the
462 publication.